CLENS — Model Evaluation

Held-out (5-fold CV) results for both models

Linear regression — 5-fold CV

Target	R^2	MAE
CO_2 (kg/kg)	0.998	0.164
Runoff (0–1)	0.996	0.00932
Plastic (0–1)	0.993	0.0143
Water (L/kg)	0.999	65.4

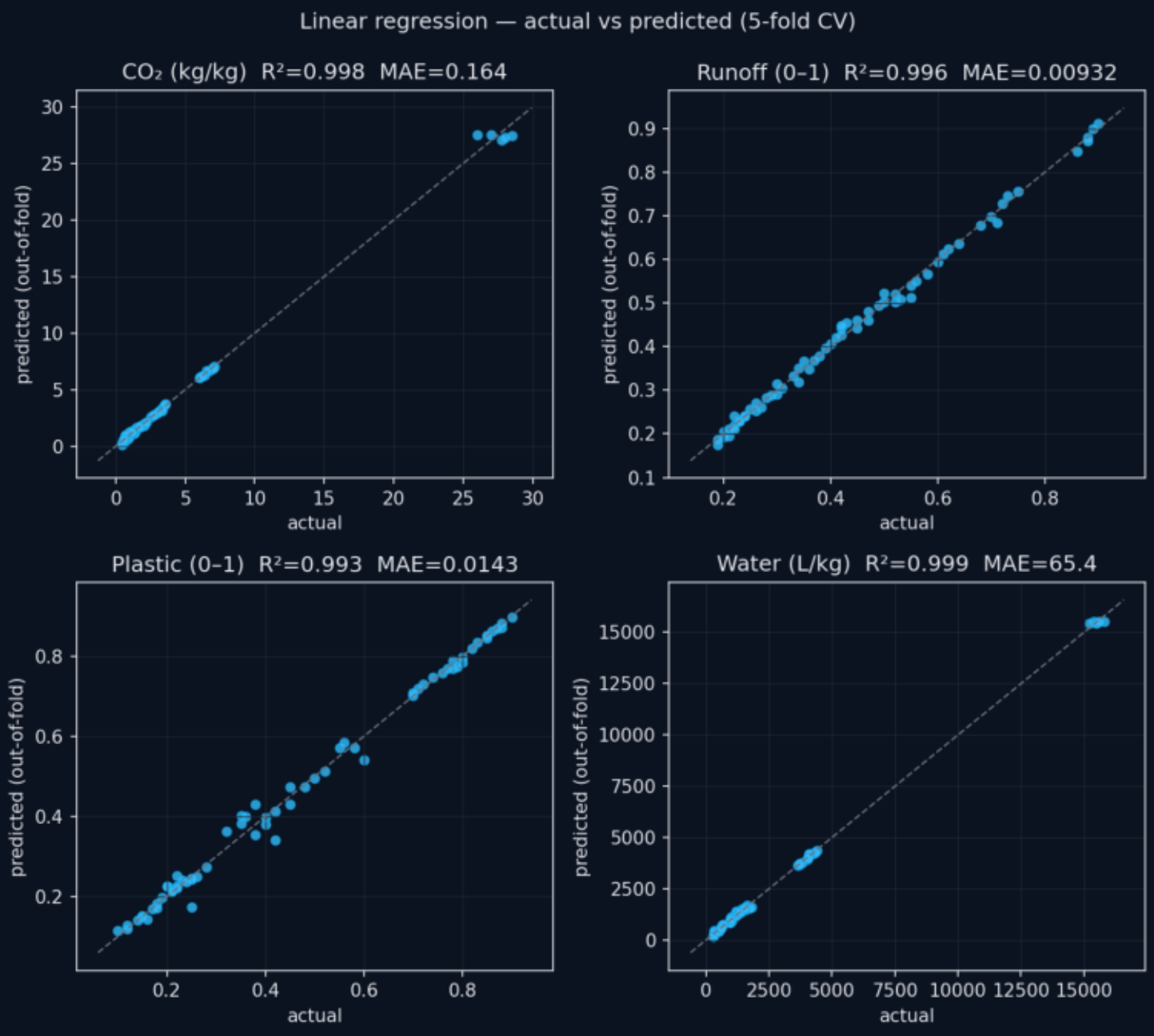
CV R^2 range: 0.993–0.999. Out-of-fold — not overfit.

Isolation Forest — 5-fold CV on synthetic CCE-style series

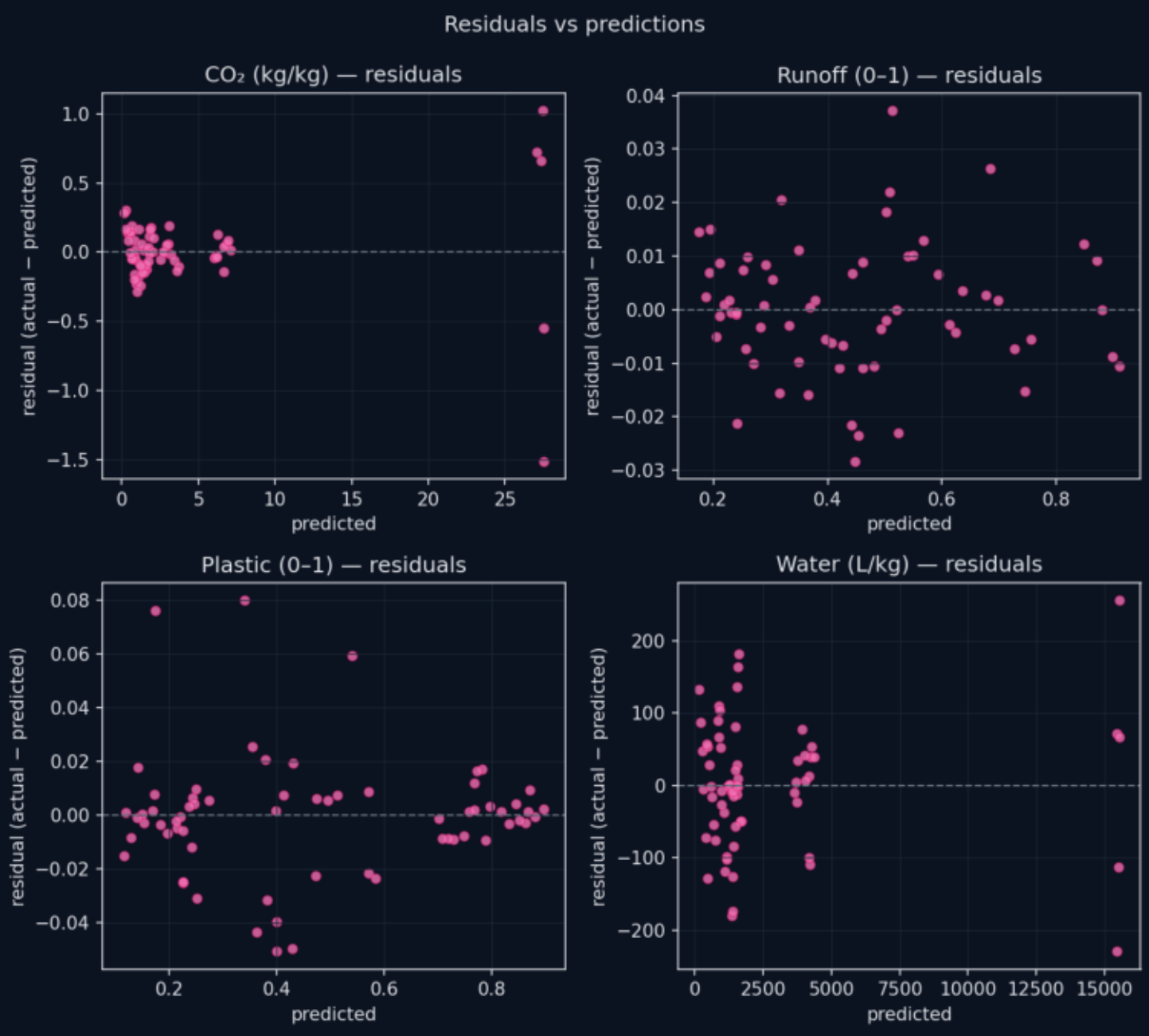
Metric	Value
ROC-AUC (OOF)	0.975
Avg Precision (OOF)	0.628
F1 (default threshold)	0.59
F1 (best threshold)	0.67
Precision @ best F1	0.50
Recall @ best F1	1.00

Out-of-fold — no metric is computed on data the model trained on.

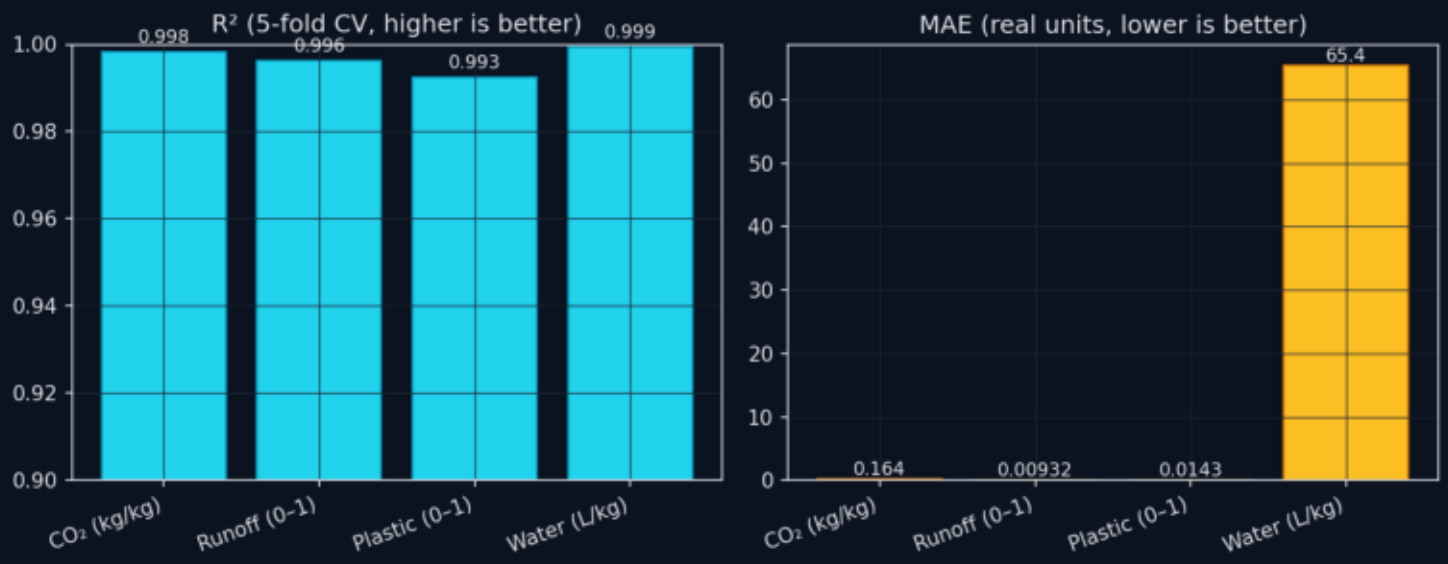
Linear regression — actual vs predicted (5-fold CV)



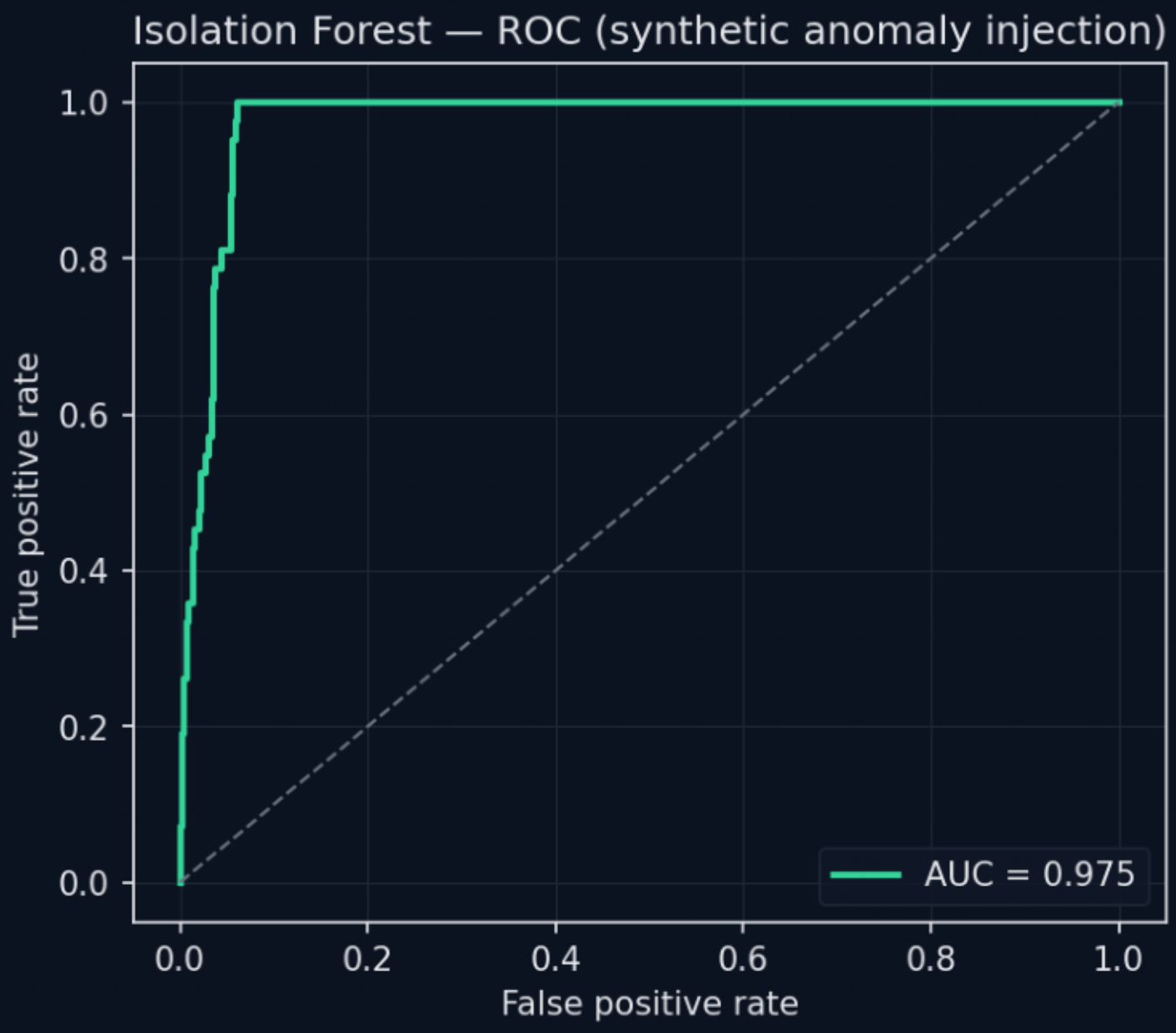
Linear regression — residuals



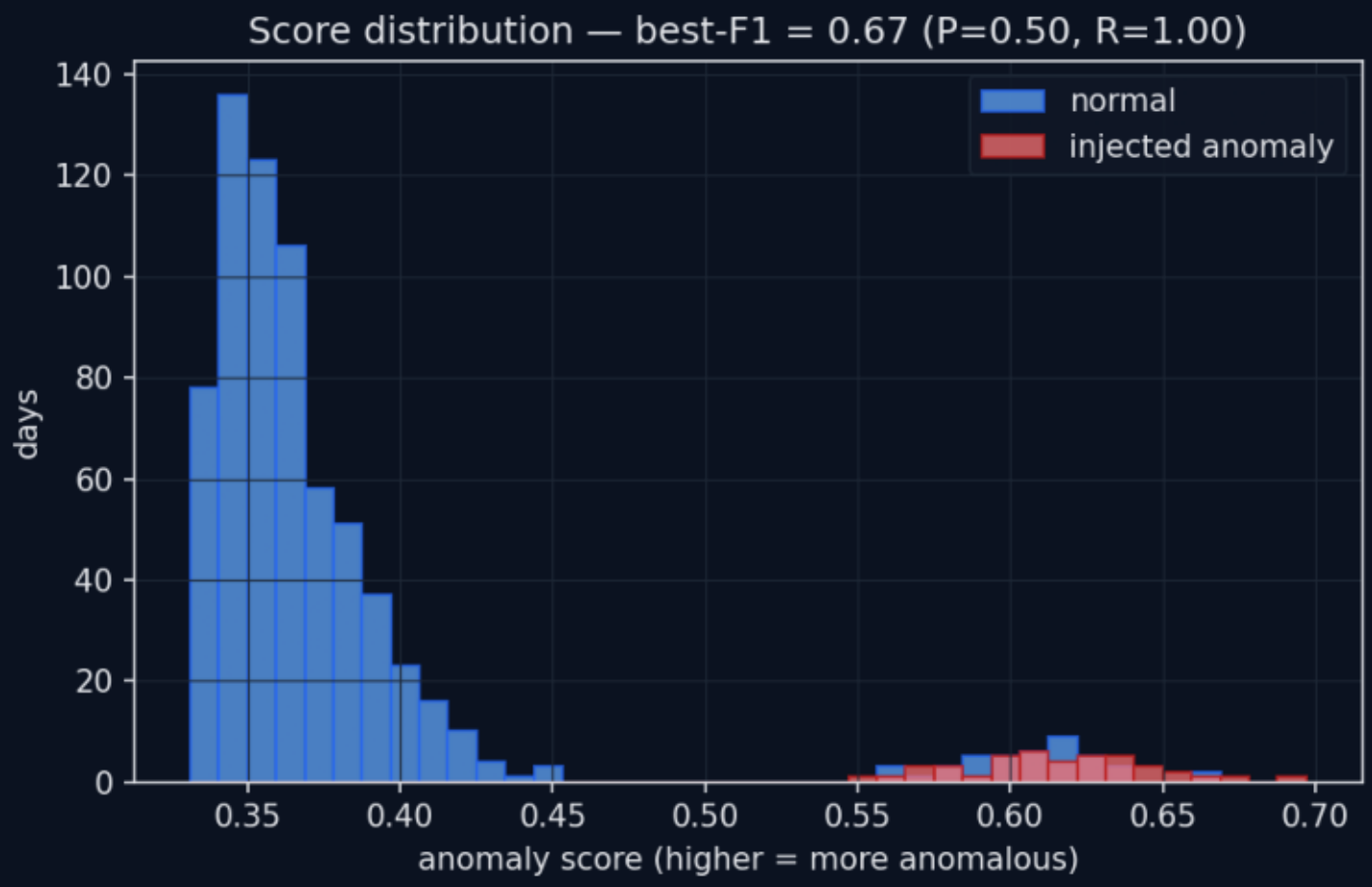
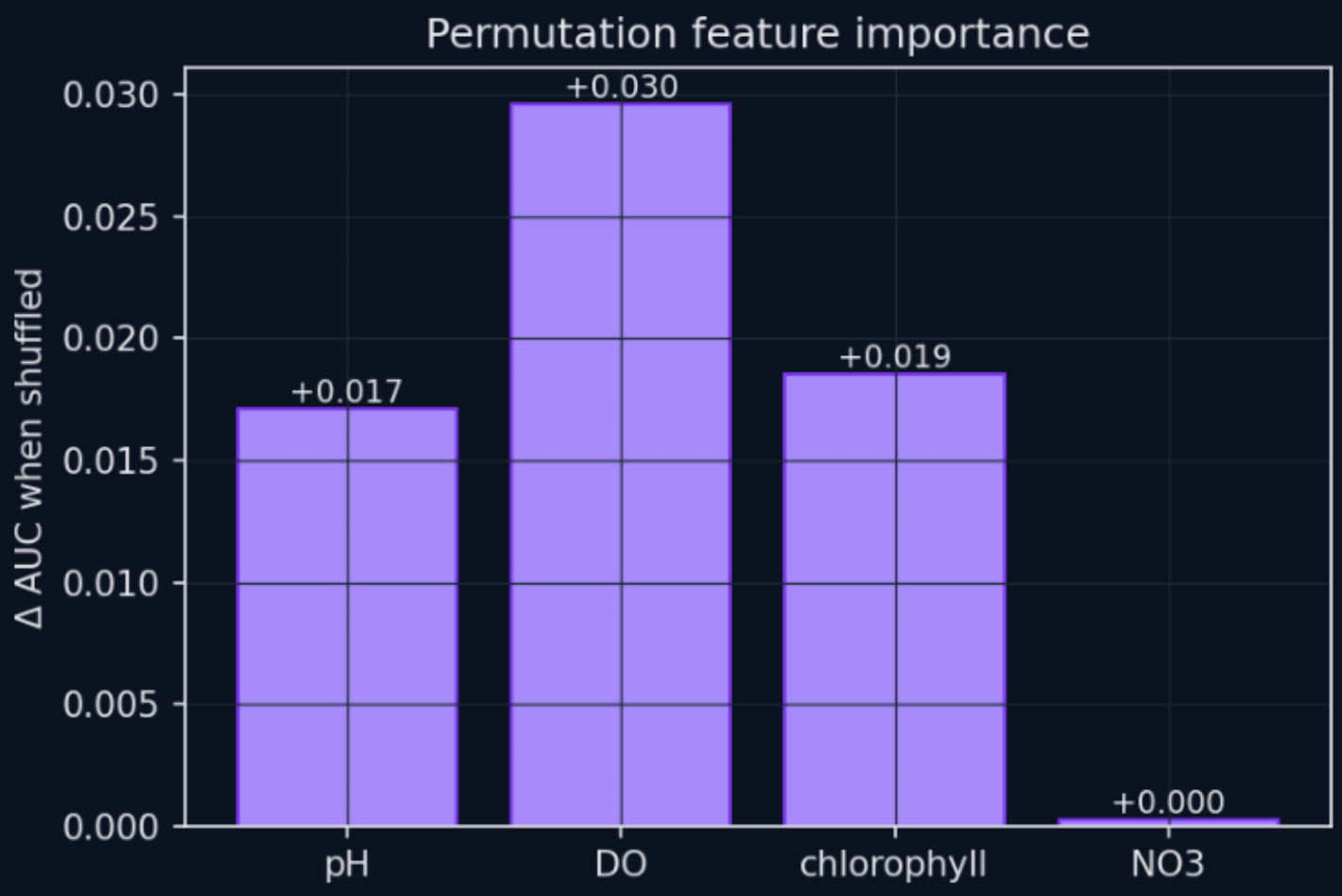
Linear regression — R² and MAE



Isolation Forest — ROC (5-fold CV, out-of-fold)



Isolation Forest — anomaly score distribution



Isolation Forest — permutation feature importance